TMN4013/TMT4703-Distributed Systems
Assignment 1

A. Overview

The purpose of this assignment is to strengthen students' understanding of multithreading, parallel processing, workload partitioning, thread synchronization, and performance evaluation.

In this assignment, each group is required to transform a sequential Java brute-force MD5 password-searching program (see Attachment A) into a Java multithreaded application. The application must dynamically create multiple threads, partition the search space among them, and coordinate their execution to identify the target password.

In addition to implementing a functional multithreaded application, students must conduct performance experiments using different numbers of threads, analyse the observed execution time, speedup, and parallel efficiency, and evaluate the influence of the experimental hardware configuration on application performance.

MD5 algorithm is a widely used cryptographic hash function producing a 128-bit hash. An MD5 hash value is typically expressed as a 32 digits hexadecimal number. For example, MD5 hash value for "football" is "37b4e2d82900d5e94b8da524fbeb33c0". An online MD5 hash generator is available at https://www.md5hashgenerator.com/

B. Requirement

Below are the requirements/functionalities that need to be demonstrated in this assignment.

a) A Java multithreading program that provides a Command Line Interface (CLI) to receive three inputs: (a) an MD5 hash value, (b) the number of threads to be created for a brute-force MD5 password search, and (c) password length. For example:

```
java MD5Search <hash> <threads> <passwordLength>
```

b) Each group will be assigned five MD5 hash values generated from 2-, 3-, 4-, 5-, and 6-character passwords. The passwords contain ASCII characters with decimal values ranging from 32 to 126, providing 95 possible characters for each character position.

   For each assigned MD5 hash, the application must search the corresponding password length specified in Appendix B. The search must include all possible combinations of the 95 ASCII characters (decimal values 32–126) for that password length.

The group must define and consistently apply a deterministic order for generating password candidates. The same candidate-generation order must be used throughout the performance experiments.

The assigned MD5 hash values will be used for the following purposes:

i. **Functional Verification:** The 2- and 3-character passwords must be successfully recovered to demonstrate correct application functionality, including password searching, multithreading, and thread coordination.

ii. **Performance Evaluation:** The 4- and 5-character passwords must be used for the performance experiments specified in Section B(f), including execution-time measurement, speedup, and parallel efficiency.

iii. **Extended Testing:** The 6-character password is provided as an optional extended test to evaluate the application under a substantially larger search space. Successful recovery of this password is not required for full marks.

The application must perform an actual brute-force search to recover each password. Hard-coded passwords or predefined hash-to-password lookup tables are not permitted.

The assigned MD5 hash values are provided in Appendix B.

The table below shows a sample of passwords with different lengths and their corresponding MD5 password.

Password	MD5 hash value
a!	a46da3640231e41e1782925a91e46103
A!1	3318d2b64514b3df4168659e5886b0a9
Abc$	e0ff1cfaacee2e8a46e8c2f95ab83cb1
Abc$9	30e1f195ab39ea0b746e9f3842cd0462

c) The number of allowed threads ranges from 1 to 10 threads.

d) The search space must be divided among the Java threads according to the number of threads specified by the user. Each thread must be assigned a non-overlapping portion of the search space, and the combined search ranges must cover the complete search space. The partitioning mechanism must also handle cases where the total search space cannot be divided equally among the threads.

The group must explain and illustrate the search-space partitioning mechanism in the MS PowerPoint presentation and pre-recorded

demonstration video. The explanation must include an example of how the search space is divided among four Java threads.

e) The searching process must stop once the corresponding password is found by any of the created threads. The application must implement an appropriate synchronization or thread-coordination mechanism to communicate the successful search result to the other threads and terminate their searching activities promptly.

The application must report the recovered password, the thread ID that found the password, and the total execution time. The main program must ensure that all worker threads have terminated before reporting the final execution time.

The group must explain and justify the selected thread synchronization or coordination mechanism in the presentation.

f) The performance of the Java multithreaded application must be evaluated using the following thread configurations: **1, 2, 4, 6, 8, and 10 threads**. For the performance experiments, each group must use its assigned 4-character and 5-character MD5 hashes. The same computer and software environment must be used throughout the experiments.

Each thread configuration must be executed at least three times. The execution time for each run must be recorded, and the median execution time must be calculated for performance comparison.

The following experimental results must be included in the MS PowerPoint presentation:

i. **Execution Time:** A line chart showing the median execution time against the number of Java threads.
ii. **Speedup:** Calculate the speedup for each thread configuration using:

$$Speedup\ (N) = \frac{T_1}{T_N}$$

where T_1 represents the median execution time using one thread and T_N represents the median execution time using (N) threads.

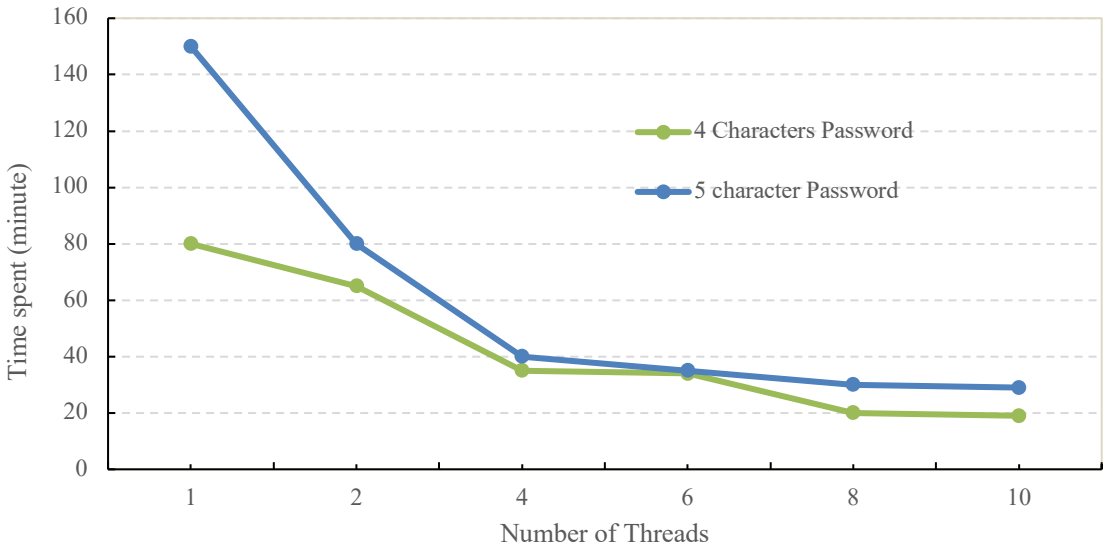
iii. **Parallel Efficiency:** Calculate the parallel efficiency using:

$$Efficiency\ (N) = \frac{Speedup(N)}{N} x100\%$$

Since the MD5 password search terminates once the target password is found, the number of password candidates examined may differ across thread configurations. Therefore, the observed speedup and parallel efficiency may be influenced by both parallel execution and early termination. Students must consider this effect when interpreting their experimental results, particularly if superlinear speedup or efficiency greater than 100% is observed.

iv. **Performance Analysis:** Explain the observed trends and discuss why increasing the number of threads may not always produce proportional performance improvements.

Separate performance tables must be provided for the 4-character and 5-character password searches. The execution-time, speedup, and efficiency results must be presented using appropriate charts.



g) The Java application must implement appropriate input validation and exception handling to prevent unexpected program termination. This must include handling invalid MD5 hash values, thread counts outside the permitted range (1–10), and missing or incorrectly formatted input arguments.

h) All performance experiments must be conducted using a single physical computer with a multicore processor. A computer with at least four physical CPU cores and 8 GB RAM is recommended.

Each group must document the hardware and software configuration used for its experiments, including:

i. Computer type (desktop or laptop).
ii. Processor manufacturer and model.
iii. Number of physical CPU cores.
iv. Number of logical processors.
v. Installed RAM.
vi. Operating system and version.
vii. Java Development Kit (JDK) version.

The hardware and software specifications must be presented in the MS PowerPoint slides using the following format.

Table 1. Experimental Hardware and Software Configuration

Component	Specification
Computer type	

Processor model	
Physical CPU cores	
Logical processors	
Processor base frequency	
Installed RAM	
Operating system	
JDK version	

Experimental conditions:

- All experiments must be performed on the same physical computer.
- The application must use the same search algorithm and workload-partitioning mechanism throughout the experiments.
- Only the number of Java threads may change between configurations.
- Computationally intensive background applications should be closed during experiments.
- A consistent power or performance setting must be maintained. Laptops should preferably be connected to AC power.
- Students must report hardware or environmental limitations that may influence their results.

Virtual machines and cloud computing instances should not be used for the formal performance experiments unless approved by the lecturer.

Students must discuss how the number of physical CPU cores and logical processors may influence the observed performance of their multithreaded application.

i) Based on the experimental results, each group must address the following questions in its MS PowerPoint presentation.

Question 1 – Performance Analysis
Analyse how increasing the number of Java threads from 1 to 10 affects the execution time of the MD5 password-searching application. Explain the observed performance trends based on the experimental results and hardware configuration.

Question 2 – Performance Evaluation

Evaluate whether increasing the number of Java threads from 8 to 10 provides a meaningful performance improvement. Justify your answer using execution time, speedup, and parallel efficiency.

Question 3 – Design Improvement

Propose one modification to improve the performance or efficiency of the multithreaded MD5 password-searching application. Explain how the proposed modification works, justify its expected benefits, and describe how its effectiveness could be evaluated.

The proposed modification, technical explanation, and justification are compulsory. However, implementation of the proposed modification is optional.

j) Each group must maintain a record of the experimental results obtained from the performance evaluation. The results must include the execution time for each of the three experimental runs, the median execution time, speedup, and parallel efficiency.

The following table format may be used.

Table 2. Experimental Performance Results

Theads	Run 1 (s)	Run 2 (s)	Run 3 (s)	Median (s)	Speedup	Efficiency (%)
1					1.00	100.00
2						
4						
6						
8						
10						

A separate table must be prepared for each password length used in the formal performance evaluation. Students must use actual measurements obtained from their Java application. All recorded experimental results must be consistent with the charts and calculations presented in the MS PowerPoint slides.

The assignment assessment will be based on the presentation/pre-recorded demonstration/presentation video, presentation slide, and Java code. Each group needs to demonstrate the application according to the functions listed above.

Grouping

This is a 5-member group assignment. Please refer to Appendix C for each group leader. The same grouping will be used for the next assignment. All the groups need to register their group members at eLeap for the assignment submissions.

C. Submission

Following are the list of items that need to be submitted:

1. Source code and executable files of the assignment.
2. An MS PowerPoint presentation that explains and includes:

a) Task distribution among the group members (who did what).
b) Detailed description of how the MD5 password-searching application is implemented using Java multithreading.
c) An experimental results file in CSV or Microsoft Excel format containing the execution times of all experimental runs, median execution times, speedup, and parallel efficiency for both the 4-character and 5-character password searches.
d) Explanation and illustration of the search-space partitioning mechanism.
e) Explanation of the thread synchronization and coordination mechanism.
f) Instructions for compiling and executing the Java application.
g) Sample output demonstrating successful password searches.
h) Hardware and software specifications of the computer used for the performance experiments (Section B(h)).

i) Experimental results, including execution time, speedup, and parallel efficiency (Sections B(f) and B(j)).
j) The recorded results must correspond to the tables and charts presented in the MS PowerPoint presentation.
k) Performance analysis and answers to the three critical analysis questions (Section B(i)).
l) Link to the pre-recorded demonstration/presentation video.

3. A maximum 15 minute presentation/pre-recorded demonstration/presentation video based on the presentation slide.
4. Both item 1, 2 and 3 need to be submitted to eLeap in compressed format with the following folder structure and naming convention. However, the naming for the source code can be arbitrary.

```
TMN4013_<NC01>_Assignment1 or TMT4703_<MC01>_Assignment 1
|
+--- Source.Code
       +---A.java
       +---B.java
+--- NC01_TMN4013_Assignment_1.pdf / +--- MC01_TMN4013_Assignment_1.pdf
+--- NC01_TMN4013_Assignment_1.ppt / +--- MC01_TMN4013_Assignment_1.ppt
+--- NC01_TMN4013_Assignment_1.xls / +--- MC01_TMN4013_Assignment_1.xls
```

5. Upload item 3, the pre-recorded demonstration/presentation video, to your cloud storage (like Google Drive, OneDrive, Dropbox) or YouTube channel and share the link to the video in the presentation slide. Do make sure that the video is unlisted if the video is uploaded to YouTube channel.

D. ***Due Date: 22 Nov. 2025 (Sunday) – before or at 2359 hours***

E. Grading (Application)

Application [10%]				
Criteria	**Unsatisfactory [0 – 2 marks]**	**Fair [3 – 5 marks]**	**Good [6 – 8 marks]**	**Excellent [9 – 10 marks]**
Multithread and password searching **Weight: 10**	• No functional multithreading implementation, or threads do not perform meaningful concurrent searches. Fewer than two compulsory passwords are recovered.	• Basic multithreading is implemented, but thread creation is restricted or the implementation contains significant functional limitations. Only two compulsory passwords are recovered.	• The application supports multiple Java threads and successfully recovers at least three compulsory passwords. Minor implementation limitations remain.	• Java threads are dynamically created according to user input (1–10 threads). All compulsory passwords are successfully recovered through actual concurrent brute-force searching, and the application operates correctly across the required configurations.
Search-space partitioning and synchronization **Weight: 10**	• Search-space partitioning is absent or incorrect. Threads duplicate substantial searching work, and synchronization or termination is not implemented correctly.	• Basic search-space partitioning and thread coordination are implemented, but overlapping ranges, incomplete coverage, or unreliable termination occur.	• Search-space partitioning is generally correct, with minimal or no overlap. Threads communicate successful discovery and terminate, although minor coordination or implementation limitations remain.	• Search space is correctly partitioned into non-overlapping ranges with complete coverage, including unequal divisions. Thread coordination is safe and reliable, other threads terminate promptly after password discovery, and the mechanism is clearly explained and justified.
CLI and error handling **Weight: 5**	• CLI is missing or does not accept the required inputs. Error handling is absent, and invalid inputs cause application failure.	• CLI accepts some required inputs, but input validation and error handling are limited. The application may terminate unexpectedly for certain invalid inputs.	• CLI accepts MD5 hash values and thread counts. The application displays the required search results and handles most invalid inputs appropriately, with minor limitations.	• CLI correctly accepts and validates MD5 hash values and thread counts (1–10). The application reliably displays the recovered password, thread ID, and execution time, with appropriate error messages and stable execution.

Experimental configuration and performance results **Weight: 10**	• Experimental hardware configuration is not documented. Performance measurements are missing, substantially incomplete, or unreliable. Required charts and calculations are absent. •	• Basic hardware specifications and experimental results are provided, but some thread configurations or repeated measurements are missing. Execution-time, speedup, or efficiency calculations contain significant errors or omissions.	• Hardware and software specifications are documented. Most required experiments are completed with repeated measurements. Median execution time, speedup, and efficiency are generally correct and presented using appropriate tables and charts, with minor omissions. •	• Complete hardware and software specifications are documented. All required thread configurations (1, 2, 4, 6, 8, 10) are tested using consistent experimental conditions, with at least three runs per configuration for both password lengths. Median execution time, speedup, and efficiency are accurately calculated and clearly presented using appropriate tables and charts.
Critical analysis and design improvement **Weight: 5**	• Little or no analysis of experimental results is provided. Performance trends are not explained, and no meaningful design improvement is proposed.	• Basic performance observations are presented, but explanations are superficial or not adequately supported by experimental results. The proposed improvement lacks justification.	• Performance trends are analysed using experimental results. The effectiveness of increasing thread counts is evaluated, and a reasonable design improvement is proposed with supporting justification.	• Comprehensive, evidence-based analysis explains the effects of thread count, hardware configuration, synchronization overhead, and early termination. The effectiveness of the 8-to-10-thread increase is critically evaluated using measured results. A technically sound improvement is proposed and justified, including how its effectiveness could be evaluated.

$$\textbf{\textit{Assignment Marks}}\ (\%) = \left(\frac{x}{40}\right) \times 10$$

x is the students total mark for this section

F. Grading *(Presentation video/Presentation)*

Presentation [10%]				
	0 - 2 Marks	**3 -5 Marks**	**6 – 8 Marks**	**9 – 10 Marks**
Technical explanation and accuracy **Weight: 10**	• Presentation is confusing or contains substantial technical inaccuracies. The multithreading implementation, search-space partitioning, and synchronization mechanisms are not adequately explained.	• Basic explanation of the multithreaded application is provided, but important technical details are missing or unclear. Limited explanation of search-space partitioning and synchronization is presented.	• Presentation clearly explains the multithreaded implementation, search-space partitioning, and synchronization mechanisms. Most technical information is accurate, with minor omissions or insufficient justification.	• Presentation provides a clear, accurate, and comprehensive explanation of the multithreaded implementation, search-space partitioning, and synchronization mechanisms. Technical decisions are well justified using appropriate diagrams, examples, and program execution evidence.
Application demonstration and explanation of results **Weight: 10**	• Application demonstration is missing or unsuccessful. Experimental results are absent, substantially incomplete, or not clearly presented.	• Basic application functionality is demonstrated, but the demonstration is incomplete or contains errors. Limited experimental results are presented, with inadequate explanation of execution time, speedup, or efficiency.	• Application functionality is demonstrated successfully using different thread configurations. Experimental hardware specifications and performance results are presented, including execution time, speedup, and efficiency, with generally clear explanations and minor omissions.	• Application functionality is successfully and systematically demonstrated, including different thread configurations, password discovery, and early termination. Experimental hardware specifications and performance results are clearly presented using appropriate tables and charts. Execution time, speedup, and efficiency are accurately interpreted and supported by the recorded measurements.

Presentation clarity and group participation **Weight: 10**	• Presentation is poorly organized and difficult to follow. Audio or visual quality significantly affects understanding. Little evidence of active participation from group members.	• Presentation has a basic structure but lacks clarity or consistency. Slides or demonstrations contain readability issues. Participation is uneven, with some members contributing minimally.	• Presentation is logically organized, with clear slides, readable diagrams, and appropriate demonstrations. Audio and visual quality are satisfactory. All group members participate and explain their assigned contributions.	• Presentation is professional, coherent, and delivered within the 15-minute limit. Slides, diagrams, and demonstrations effectively support technical explanations. Audio and visual quality are clear, and all group members participate meaningfully, demonstrating understanding of their respective contributions.

$$\textbf{\textit{Presentation}}\ (\%) = \left(\frac{x}{30}\right) \times 10$$ where x **is the students total mark for this section.**

Appendix B: List of Assigned MD5 Hash Value

Group	MD5 Hash Value				
	2-character password	3-character password	4-character password	5-character password	6-character password
NC1	d844853b604c0c55996b9c26d035662b	950ab1e1d543986f403944b898186680	f46f9ff502df0c46e07ede336794fd9c	45ac8bcb439d803b37899f792f4fe8b5	ed636e491e306244cbc61684951d983e
NC2	32d379e4fba692d8e3e76a6918228806	f7c5d53f3c962fc8b9d1ac36513a85ed	596323795c7813d7313d4751dc43c37b	9c629ea7e76fcc76066aab6db7ca8744	2f1ba477bd9458400a32e40ee0a1ce15
NC3	263a6fee6029b304bd1cf5ce0a782c6b	77aaa4dcce557f10d97b3ed037de33fb	9d64f0e38b080d131c1a27140df4e13b	e76b29d2dfffb1a327d49a797d34c8a7	f7808b86b6e53a97313f24a3619fdc95
NC4	5665da3452d54dcd10aee61c566c807f	40253fd9cc80ec22004edd8f6b5daa2a	6f2dbe20c2d22871855f8b52d40bc4eb	5c6e9b6db68fed0b85120016d7541a6d	28ad6d07c0086511cb10809644423d36
NC5	f83ba497477f1b653cb3af0bcd5bd7f9	836179de02941a3299653774fe29b76f	6683526e0d865149c9ca4dd3cd0d1889	3d710e5ecefb3d4d2117004272c8079e	35018770ca382f50de535bc81594d645
NC6	d66e695936c08d3cb4c28f33e089582f	659b027e3c7e484523bea84b97b539cf	24b980a368d2b142eb954359bdd9a890	797182a116009efdd5ee7d974b65b8fa	014f421fbf8cd29a8a5fc0ed1ec57d59
NC7	173ac641a29f57f5828c51c44d3ebe4b	6ad3c00136138fad3429ef1b02a51ead	ed1068feca153e029c8f2bc54811e5c0	c91272842728fde36a7209fbbcb52447	3cfaac0d2fedfc3d133c6d47bfb91e20
NC8	2436f23b98933a71439259c4bca8674b	7e612430dbeae4bd8bf812f8ddeed19d	e8d2fcc7255da33e359e0fdca94156ab	5a28e44b8766593a4be4905a66052d71	60dd3b2711a98621a229f991d2ac3a07
MC1	18fcd73eae8f118b0bf5f4cf5d1d766e	86bc9b110b56b40d35abbd0430f6e21d	13b4960c76af483135ce8f7de1b77a7a	fc56a489a60822bba4beada24269871f	0ab53a731f0055dcf776f76aca0b7495
MC2	dd8ae9e1d1efaa09b4094bcf3695ba4f	b3f5a7c0fccd62a7cfc4cdec29f3b346	36fefaf850a41b600d1b275f4ffb260b	06977ec387f78f3fcd62341314335e37	2969eabad2018a6972b41cda71f8d78f
MC3	25f56e214944dbe4cf864367a72fce93	bcd5a6d413dd4ea474d58ea51a824e76	ab201fbd2b5de7c81167f70c1479047a	29ad5967e2c85051008d2e16e19e2db6	8829a18344c637b03ebda76df47a7e66
MC4	67ef9f5f1d2cadd61620914ed32a2a19	cd7cca56f9d4a7d61c1430ea0364e43c	6a3b3c12eb8c8ca1e51cc3e32855e9c0	9031d22f97fae8eef8a331c11bae5ba4	cf577eae44eba7576499860c74ee45be
MC5	43b508c13b3c575bcf2556558674fe10	9b5d99c0cf2c85e970bf7014f9ce8134	32d3971fff9e081490c614319d63d594	0ca0c572ef0128c5262dfb7ef0dc25c2	0fdb30ce0777f4cc1585e749e29cae6f
MC6	a7897b68a0645cd54c92fcf496934eb9	b7d785d1868741c3fe48d7518950968b	f0a72273c3112e3721b632cb795349ea	faf4f914b73507f05d676663c3c0cee8	f60a52fb3a0513b29e6d785c3b42dd44
MC7	a3f390d88e4c41f2747bfa2f1b5f87db	bc4952663d3c03ddeb731c2e0e4ddaf6	71c2d7de74c606e57378b878ad0b3465	714b40bc5d85ee388db4d9a8d5ce7e06	2ef7b87498920ecb636285cd51df971e
MC8	bc671e5f5b664419dedab3ea70b22f2c	aede4f407826188b785e51f07b99b129	fc18102f850d92a7a29eabe88d9d7a1a	764a6db7d0bb9f5acdc80ebf4ce2d14a	d1a4fa4914ba1285c1b73c51681515c1
MC9	55b87b4d42da5c9344a6c3e5f6b70d95	511b388b42e6cf192691122ef4f577ec	32d32f1ecdd66feda52ddc77700a695c	164775a550545dd43f4dbe18716a5e7f	814cfb2cea0ed9d0e416f8b80ffd5567
MC10	fa5199613525eb9f67ae6d8f3ca8db14	078aab7ae8c1a0df339ac8525790d5e1	5e1fec96a62844d1b842bb4f37909b84	8b1dde15a62052e26670d779737751d8	4b4191a67c0fe1f87b66d21eb736414c
MC11	22c54b590daed50bbf6cb33e45e960d3	9ff3d81acb5458fe8150566689e23f16	10fa797445102410e9c32dc75e7dc8f2	0410600fadd0ad0259e23812a7fccee9	ccd9474f7e023db72b28c0424516d31c
MC12	532a7b8e0328a8d05a8e6258b28b9a36	9ff3d81acb5458fe8150566689e23f16	efaadd88b128766b3a84aa94c166b05d	f5a826b78b0d1d59871656cbf2fbda12	4212592635e037c6ab28a075906c38ba

Appendix C:

Network Computing Group Leader

1	Group Leader 1	AMIRUL IMAN BIN AHMAD RAZLAN
2	Group Leader 2	CALVIN KULING ANAK LANGGONG
3	Group Leader 3	FEMILIA LUSA ANAK JIHED
4	Group Leader 4	JULIAN WALIS ANAK DENSI
5	Group Leader 5	MARK ANTHONY ANAK CHRISTOPHER
6	Group Leader 6	NUR IMANINA BINTI CHIN
7	Group Leader 7	SARVIN RAMEN A/L REGU RAMEN
8	Group Leader 8	WHITELLMINA VATHA ANAK PATRICK LAYONG

Multimedia Computing Group Leader

1	Group Leader 1	ANGEL BONG XIN TZE
2	Group Leader 2	CHLOE TAN YOU TIAN
3	Group Leader 3	EVAN ARNOLD JUINIS
4	Group Leader 4	IVY JONG ZI ENG
5	Group Leader 5	LEE WEN RONG
6	Group Leader 6	MUHAMAD SYAZMI RAFIZEE BIN RASID
7	Group Leader 7	NATALIE ANAK VINCENT
8	Group Leader 8	ODILON CORINO ANAK JUNIT
9	Group Leader 9	SAMUEL AGAL DICKY
10	Group Leader 10	TEOFILUS HEZRON BLAUN ANAK JANA
11	Group Leader 11	WAN CAIREL AIMAN BIN WAN HABIB (6 members)